

Paper Manager pack — UMI (Nov 16 · Data)

Paper: Universal Manipulation Interface: In-The-Wild Robot Teaching Without In-The-Wild Robots
arXiv: <https://arxiv.org/abs/2402.10329> · [project](#) · [code \(MIT\)](#) · **RSS 2024** (Best Systems Paper Award Finalist — project page)
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PDF: `/workspace/papers/umi-2402.10329.pdf` (= `/workspace/umi-quiet/paper.pdf`) · 18 pp · arXiv 2402.10329v3 (6 Mar 2024)
Session: Nov 16 · due 9:30 AM · Type: Data
Role weights: Empiricist=1 (Kanta — LONGEST / PRIMARY) · Scientific Reviewer=4 (substantial) · Visionary=2 (substantial) · Stakeholder / Archaeologist seats empty — still all five produced
Kanta role: Empiricist — deepen env/run/GPU/scripts + exact reproduce commands; produce all five
Themes: handheld gripper · in-the-wild teaching without robots on-site · relative SE(3) · latency matching · wrist fisheye · Diffusion Policy chunks
Sources: Summarizer + Ablation Bot + Senior Research (`/workspace/umi-quiet/DELIVERABLE.md`) + `paper_notes.md` / `env_outline/`
Sibling links:* `/workspace/diffusion-policy-quiet/` · `/workspace/act-aloha-quiet/` · `/workspace/droid-quiet/` · `/workspace/oxe-quiet/` · `/workspace/dart-quiet/` · `/workspace/pi05-quiet/` · `/workspace/openpi-comet-smoke/` · `/workspace/openvla-quiet/`

1. One-liner

In-the-wild teaching without in-the-wild robots — relative EE + latency honesty turns **human logistics** into robot-ready demos; complementary to **DROID** (robot-in-wild) and **ALOHA** (lab teleop).
Contrast poles:

Pole	Who collects	Where	Action interface
UMI	Human + handheld gripper	Anywhere (cafe, fountain, home)	Relative SE(3) EE + wrist fisheye + latency match
DROID	Robot teleop (Franka fleet)	564 real scenes / 52 buildings	Absolute EE @ 15 Hz (paper DP)
ALOHA/ACT	Lab leader-follower	Lab tables	Joint-space, embodiment-specific

2. Summary (exec overview + key numbers)

UMI is a **data collection + policy learning** framework: collect demos anywhere with a portable ~\$371 GoPro + 3D-printed parallel-jaw gripper (no robot during collection), recover metric 6DoF EE via **ORB-SLAM3 + GoPro IMU**, train **Diffusion Policy** on **relative SE(3) action chunks** ($T_a=6$ @ 10 Hz, or 20 Hz tossing), deploy zero-shot on **UR5 / Franka FR2** with **inference-time latency matching**.
Why Data session: Interface design (FoV, IMU-SLAM, latency, relative actions, multimodal DP) — not just “more scenes” — unlocks **action diversity** (toss, fold, wash) from handheld demos that prior handheld systems limited to grasp/pick-place.

Axis	Number / claim
Cup lab (UR5)	20/20 = 100%
Cup → Franka (same ckpt)	18/20 = 90% (2 joint-limit)
Absolute / Delta / Rel action	25% / 80% / 100%
No fisheye (69° crop)	55%
Toss w/ latency / w/o	105/120 = 87.5% / 69/120 = 57.5%
Cloth fold / no inter-grip	70% / 30%
Dish wash ViT / ResNet	70% / 0%
Wild cup OOD	1400 demos / 30 loc / 12 person-h → 43/60 = 71.7%; narrow-only 0%
SLAM ATE / RPE	6.1 mm / 3.5° · inter-grip 10.1 mm / 0.8°
Throughput vs teleop	>3× spacemouse (cup); toss demos teleop 0 in 15 min
BoM	Gripper \$73 + GoPro \$298; mass 780 g
DP chunks	Ta=6 @ 10 Hz (toss 20 Hz); DDIM 50/16
Example exec latency	Arm ~100 ms / gripper ~120 ms (Fig. 5 — measure per install)
GPUs (Tab. A1)	Lab 4×A10g (dish 8×); wild 8×A100 — wall-clock hours NOT in paper

Must-cite tattoo: cup 100%/90% · toss 87.5% (no lat 57.5%) · wild 1400/30 → 71.7% vs narrow 0% · rel 100% vs abs 25% · fisheye 100% vs 55° 55% · $\pi_{0.5}$ / GR00T CLEAR cite UMI.

3. Keywords

UMI; handheld gripper; in-the-wild teaching; GoPro; fisheye wrist camera; side mirrors; ORB-SLAM3; visual-inertial SLAM; relative trajectory action; latency matching; Diffusion Policy; action chunking; DDIM; Ta=6; bimanual inter-gripper proprioception; continuous gripper width; soft TPU fingers; zero-shot transfer; RSS 2024; DROID contrast; ALOHA contrast; $\pi_{0.5}$; GR00T-N1; BEHAVIOR handheld recipe

4. Deep dive — Motion / latency / relative EE (PRIORITY)

Callout: UMI is where DP’s chunk control loop meets a deployable handheld data interface. Relative EE + latency matching are the motion-control upgrades that make wild human demos robot-ready.

Chunking / horizons (Tab. A1 — CLEAR)

Knob	Paper choice	Implication
Family	Diffusion Policy (DDIM)	Multimodal human modes (cup CW/CCW)
Action horizon Ta	6 all tasks	Chunk length
Control freq	10 Hz (cup/fold/wash/wild); 20 Hz (toss)	Commit ≈ 0.6 s @10 Hz; 0.3 s @20 Hz
Obs	I-To ∈ {1,2}; P-To = 2	Short visual + relative proprio history ≈ velocity
DDIM	50 train / 16 infer	Leave room inside control period after matching
Speed scale	0.5× quasi-static; 1.0× toss	Smooth tracking vs release velocity
Action alphabet	Relative SE(3) traj w.r.t. EE at chunk start (t_0) + continuous gripper width	Not world-abs; not delta-to-previous

Relative vs Absolute vs Delta (Fig. 6 / PD2.1):
 For a chunk starting at (t_0), each action is the desired pose at (t) **relative to EE pose at (t_0)**. Absolute needs SLAM ↔ base calib (brittle → 25%). Delta accumulates step error → 80%. Relative flat under calib noise → 100%. Also enables **move robot base mid-episode** (Fig. 10).

Latency matching (PD1) — CRITICAL for dynamic

Fig. 5 illustrative deploy latencies: Arm exec ≈ 100 ms, Gripper exec ≈ 120 ms (examples — measure per install; App. A).

- 1. Measure camera / proprio / gripper / arm latencies (QR rolling clock for cam; ≈½ RTT if no HW stamp; cross-corr for exec).
- 2. Align streams to **highest-latency** (usually camera); downsample RGB to 10–20 Hz; interpolate proprio/gripper to (t_{obs}).
- 3. Predict action chunk from last obs; **discard outdated** actions covering obs+infer+exec delay; send remaining **ahead** by measured exec latency.

Toss moral: with matching 105/120 = 87.5%; disable (set all latencies → 0) → 69/120 = 57.5% (–30 pp). Jitter barely hurts grasp; hurts toss velocity + gripper–arm release timing.

Wrist cams / FoV / mirrors

- **Sole** observation modality (no external cam in policy).
- **155° fisheye** raw (no undistort) — crop → 69° → 55%.
- Side mirrors = implicit stereo; must **digitally reflect + swap L/R** (raw mirrors **hurt**: 85% < no-mirror 90%).

Long-horizon / bimanual / mobile hooks

Hook	Number	Control use
Dish wash 7-step	70% ViT; ResNet 0%	Stage-wise SR; recovery demos (ketchup re-dirty)
Cloth fold	70%; no inter-grip 30%	Dual-arm sync via relative inter-gripper proprio
Base shift mid-rollout	Fig. 10 qualitative	Relative EE ⇒ mobile-manip ready if objects in reach
Soft TPU fingers	continuous width	Contact-rich / faucet / toss release

Syllabus takeaway for Kanta: Treat **latency calib** as a **controller parameter**; ablate with horizon (H-commit). Link DP Push-T morals (~0.5–1 s commit) — UMI toss wants shorter or better-matched.

5. Deep dive — Collection methodology (handheld scaling) (PRIMARY DATA CONTRIBUTION)

FLAG: UMI's scientific product is the **portable teaching interface** (hardware HD1–HD6 + policy PD1–PD2), not a new generative head. Collection **without robots on-site** is the Data-session spine.

Hardware BoM (CLEAR)

Spec	Value
Mass / envelope	780 g · L310×W175×H210 mm
Finger stroke	80 mm; soft 95A TPU ribs (series-elastic via width)
Cost	Printed gripper \$73 + GoPro+acc \$298
Sole sensor	Wrist GoPro Hero 9 + 155° fisheye + built-in IMU (GPMF)
Mirrors	Physical side mirrors in peripheral FoV
Gripper width	ArUco/fiducial continuous (not binary)
Robot twin	Same finger/cam geometry; Schunk WSG-50; Media Mod → Elgato HD60X UVC
Arms validated	UR5 + Franka Emika FR2 (custom 90° WSG mount)

Pipeline (App. B)

- 1. Optional bimanual QR clock sync ($\pm 1/60$ s) + gripper width cal.
- 2. **Map-then-localize**: ~1 min scene mapping video → demos relocalized into shared map (enables inter-gripper pose). Optional fiducials **only during mapping**.
- 3. Repeated demos (one MP4 / episode) → upload → **single script** → SLAM + fiducials → Diffusion Policy zarr.
- 4. **Kinematic filter**: drop demos infeasible for target robot base/DoF before BC.

Throughput (§VII / Fig. 11) — same operator, 15 min

Task	Bare hand	UMI	Spacemouse teleop
Cup arrange	fastest	~48% of hand; >3× teleop (111 vs 35 CPH)	slowest
Dynamic toss	fastest	~64% of hand (149 CPH)	0 successful demos

In-the-wild scaling (Fig. 9)

- 1400 demos / 30 locations / 3 demonstrators / 12 person-hours / 15 train cup styles.
- Unseen cafe + fountain: 43/60 = 71.7% (train cups 28/40=70%; unseen cups 15/20=75%).
- Narrow-domain-only + same CLIP ViT family: 0% — “doesn’t even move toward the cup.”
- **Moral**: scene diversity in **hand-held** demos is the generalization knob (sibling to DROID scene cards; opposite *who* goes into the wild).

Open data / code

- Example session + `cup_in_the_wild.zarr.zip` + pretrained `cup_wild_vit_l` on `real.stanford.edu/umi/`
- SLAM fork: `cheng-chi/ORB_SLAM3` + Docker `chicheng/orb_slam3`
- Deploy forks: `umi-on-legs`, `umi-arx` (ARX X5)

Do not equate UMI collection scale with DROID 76k — different axis (action-rich human-handheld vs robot-in-wild volume).

6. Role-ready sections (all five)

6.1 Empiricist — WEIGHT 1 / LONGEST (Kanta PRIMARY)

Seat charge: Course smoke ablations that stress **relative action · latency · embodiment transfer · policy-from-UMI-demos**, ranked by GPU/RAM/TIME honesty. Prefer offline/proxy metrics before real-robot eval. This section + `env_outline/` is the pack’s richest deliverable.

Claim under measurement

Can a GoPro+printed gripper + IMU-SLAM + latency-matched relative-trajectory Diffusion Policy transfer dynamic/bimanual/long-horizon human demos to UR5/Franka with reproducible directional numbers on proxies?

Hypotheses (falsifiable — from DELIVERABLE)

- 1. **H-rel**: On identical demos, **relative traj** → lower held-out SE(3) err **and** higher proxy success than **absolute** (calib-sensitive) and $\geq$ **delta** (accumulates). Paper cup: 100 / 25 / 80%.
- 2. **H-lat-dyn**: Disabling latency matching drops **sharply on dynamic** toss-proxies (paper **–30 pp**) and mildly on quasi-static grasp.
- 3. **H-lat-curve**: U-shape / cliff — under- and over-compensation (wrong signed latency) both hurt velocity tracking more than position reaching.

4. **H-emb**: EE-relative + wrist-cam policy transfers across kinematic models **iff** ingest applied kin-filter and wrist FoV matched; joint-limit fails dominate residual (paper Franka 2/20).
5. **H-dp-umi**: Short DP smoke on relative chunks fits multimodal human modes better than MSE-MLP on same shard (link </workspace/diffusion-policy-quiet/>).
6. **H-bimanual**: Without **inter-gripper relative proprio**, dual-arm stage sync fails even if per-arm BC loss looks fine (paper **70** → **30%**).
7. **H-wild**: Holding N fixed, **multi-scene hand-held** mix beats single-scene on scene-OOD cup proxy (Fig. 9; sibling DROID).
8. **H-commit**: Effective commit seconds ($T_a \setminus \Delta t$) should land near DP Push-T morals ($\sim 0.5\text{--}1$ s) for quasi-static and **shorter or better latency-matched** for toss.

Falsifiers: $\text{abs} \geq \text{rel}$ under perfect synthetic calib; latency matching noop on toss-proxy; transfer works without kin-filter; ResNet matches ViT on dish-proxy with matched steps.

Prioritized ablation matrix (Ablation Bot + DELIVERABLE)

Pri	Experiment	Paper hook	Smoke	Est. GPU-h
P0	Relative vs Absolute vs Delta	Cup: Rel 100% , Abs 25% , Delta 80%	Y	2–8
P0	Latency matching on/off (+ sweep)	Toss: 87.5% vs 57.5%	Y	1–4 (eval)
P0	Demo ingest (SLAM → rel chunks + kin filter)	ATE 6.1 mm / 3.5°	Y	CPU <1
P1	Policy train smoke (DP on tiny UMI-like shard)	Fig. 5 interface	Y	4–12
P1	Embodiment transfer stub (IK/feasibility)	Franka 90%	Partial	1–3
P1	Inter-gripper relative proprio on/off	Cloth 70% vs 30%	Partial	4–10
P2	Fisheye / FoV crop	No fisheye 55%	Partial	4–10
P2	Side-mirror digital reflect	raw 85% → reflect 100%	N	—
P2	ViT-CLIP FT vs ResNet scratch	Dish 70% vs 0%	Partial	8–20
P3	Wild mix vs narrow-only OOD	Fig. 9: 71.7% vs 0%	Partial	8–24
Defer	Full real 4-task 20-trial protocol	§V	N	50–200+

Default course smoke order:

demo_ingest_smoke → relative_action_ablation × {rel, abs, delta} → latency_sweep × {0, match, 2×match} → policy_train_smoke (rel only) → embodiment_transfer_stub.

Exact reproduce stack (CLEAR from GitHub README / summary)

```
# Env (official UMI)
# Ubuntu 22.04; system: libosmesa6-dev libgl1-mesa-glx libglfw3 patchelf
git clone https://github.com/real-stanford/universal_manipulation_interface
cd universal_manipulation_interface
mamba env create -f conda_environment.yaml
conda activate umi

# SLAM (Docker fork)
# cheng-chi/ORB_SLAM3 · image chicheng/orb_slam3
python run_slam_pipeline.py <session_dir>

# Dataset build
python scripts_slam_pipeline/07_generate_replay_buffer.py \
  -o ../dataset.zarr.zip <session>

# Train – 1xGPU (RTX 3090 24GB tested)
python train.py \
  --config-name=train_diffusion_unet_timm_umi_workspace \
  task.dataset_path=...

# Train – multi-GPU
accelerate --num_processes <n> train.py --config-name=...

# Wild cup data + ckpt (public)
wget ../cup_in_the_wild.zarr.zip
wget ../cup_wild_vit_l_limg.ckpt

# Real-robot eval (needs hardware)
python eval_real.py \
  --robot_config=example/eval_robots_config.yaml \
  -i <ckpt> -o <outdir>
# SpaceMouse; C=start / S=stop
```

Course scout env (quiet outlines — preferred for class smokes):

```
export UMI_WORK_ROOT=$HOME/src
export UMI_DATA_ROOT=$HOME/data/umi_subsets # EDIT
cd /workspace/umi-quiet
bash env_outline/setup_mamba.sh
# mamba activate umi-scout
mkdir -p logs
# EDIT partition/account in slurm_*.sh – outlines only; do NOT sbatch from quiet box
# Pin official UMI + DP commits → logs/pins.txt
```

Run cards (RC0–RC4)

RC0 — demo_ingest_smoke (CPU gate)

Field	Spec
Goal	Prove demo → obs/action tensors
Input	Public UMI sample or synthetic SE(3)+gripper+RGB stubs (10–50 eps)
Steps	decode → SLAM/stub poses → relative chunks → kin-filter UR5-like & Franka-like → zarr/hdf5 + manifest
Pass	Schema OK; 0 NaNs; relative chunk starts at Identity; filter report; wall <30 min CPU
Fail	Dropped frames unaligned; abs poses leaked into “rel”; filter removes >50% without reason
Script	env_outline/slurm_demo_ingest_smoke.sh

RC1 — relative_action_ablation

Field	Spec
Goal	Rel vs Abs vs Delta on same shards
Train	DP or light temporal CNN; 1 seed ; 2k–10k steps
Axes	<code>action_repr</code> $\in$ { <code>relative_traj</code> , <code>absolute</code> , <code>delta</code> }; fix <code>obs/batch/lr/horizon</code>
ID	Held-out action MSE / rotation geodesic
OOD	SE(3) origin noise / base-frame jitter at eval only (stress abs)
Expected	Abs explodes under jitter; Delta drifts; Rel flat — ranking matches cup 100/80/25
Budget	1 GPU , ~2–8 h for 3 runs
Script	<code>env_outline/slurm_relative_action_ablation.sh</code> (array 0–2)

RC2 — latency_sweep

Field	Spec
Goal	Reproduce toss moral without full robot if needed
Setup	Fixed ckpt from RC1-rel or teacher traj; eval-only
Grid	<code>lat_mode</code> $\in$ { <code>zero</code> , <code>measured_match</code> , <code>0.5×</code> , <code>2×</code> , <code>sign_flip</code> }; optional obs delay (<code>d\in{0,1,2,4}</code>)
Metrics	Velocity-profile MSE on toss phase; release timing; jerk; proxy bin success if env exists
Expected	<code>measured_match</code> best; <code>zero</code> $\approx$ -30 pp moral; <code>sign_flip</code> worst
Budget	$\leq$ 1–4 GPU-h (mostly infer)
Script	<code>env_outline/slurm_latency_sweep.sh</code>

RC3 — policy_train_smoke

Field	Spec
Goal	End-to-end short DP on UMI-relative data
Config sketch	<code>To=2</code> , <code>Tp$\approx$16</code> , <code>Ta$\approx$6–8</code> (EDIT pin from UMI repo); DDIM 10–16; batch 32–64; img 224 or 128; relative EE 6D/9D + gripper
Steps	Scout 5k–20k (not paper full); early-stop on val action err
Pass	Finite grads; VRAM logged; val err < train $\times$ 3; $\geq$ 1 qualitative rollout gif
Fail	OOM $\rightarrow$ drop batch/res; NaN quat $\rightarrow$ use 6D rot
Budget	4–12 GPU-h
Script	<code>env_outline/slurm_policy_train_smoke.sh</code>

RC4 — embodiment_transfer_stub

Field	Spec
Goal	Test transfer without claiming Franka 90%
Method	RC3 chunks $\rightarrow$ FK/IK for robot A vs B; count joint-limit / singularity / collide
Pass	Feasibility A=B within ~10 pp when kin-filter used; gap widens when filter off
Budget	CPU / $<$ 1–3 GPU-h if replaying vision
Script	<code>env_outline/slurm_embodiment_transfer_stub.sh</code>

Exact demo counts & success (exam copy)

Task	Demos	Eval	UMI success	Key ablation
Cup lab	305	20	100% UR5 / 90% FR2	Abs 25%; no fisheye 55%; delta 80%
Toss	280	120 obj	87.5%	No latency 57.5%
Cloth fold	250	20	70%	No inter-grip 30%
Dish wash	258	20	70%	ResNet 0%
Cup wild	1400 / 30 loc	60	71.7%	Narrow-only 0%
SLAM MoCap	14 tasks	—	ATE 6.1mm/3.5°	RPE 10.1mm/0.8°

Tab. A1 train budget (cite epochs/GPUs only — no invented wall-clock hours)

Task	Ta	Freq	Speed	Vision	Epochs	Batch	GPUs
Cup lab	6	10	0.5×	ViT-B/16 CLIP	250	512	4×A10g
Toss	6	20	1.0×	ResNet-34	350	1024	4×A10g
Cloth	6	10	0.5×	ResNet-34	100	1024	4×A10g
Dish	6	10	0.5×	ViT-B/16 CLIP	90	224	8×A10g
Cup wild	6	10	0.5×	ViT-L/14 CLIP	50	512	8×A100

All: DDIM 50/16; D-Lr 3e-4; V-Lr 3e-5 (CLIP) or 3e-4 (ResNet).

Latency calibration recipe (App. A)

1. Camera: film rolling QR of system clock $\rightarrow (l_{\text{cam}} = t_{\text{recv}} - t_{\text{display}}) - l_{\text{display}}$.
2. Proprio: HW stamp delta or $\approx \frac{1}{2}$ RTT.
3. Gripper/arm exec: cross-correlate commanded vs measured $\rightarrow (l_{\text{action}} = l_{\text{e2e}} - l_{\text{obs}})$.
4. At run: sync to cam; drop stale; command ahead by (l_{action}) .

Expected ID / OOD proxies (course pass)

Proxy	ID	OOD / stress	Pass bar
Action repr	Clean held-out	Origin / base-frame jitter	Rel best under jitter; Abs collapses
Latency	Matched delay=calib	Delay $\neq$ calib, dynamic subset	Match > zero on velocity MSE
Policy smoke	Train scene val	Held-out scene folder	Loss↓; no NaN; gif OK
Embodiment	Robot A IK OK	Robot B same chunks	Kin-filter on $\Rightarrow$ transfer gap small
Wild mix (stretch)	Multi-scene train	Unseen scene	Multi-scene > single-scene @ matched N

Not required for pass: paper 100%/87.5%/70% real-robot rates, 3 seeds, full 305+280+250+258 retrain, physical tossing bins.

Failure modes cheat-sheet

Symptom	Likely cause	Next action
Abs $\approx$ Rel on clean data	Missing origin jitter OOD	Add eval noise (H-rel stress)
Rel worse than Delta	Relative transform bug	Unit-test Identity@t0; compare Fig. 6
Latency noop	Sweeping only position tasks	Restrict to high-speed segments
OOM mid-train	ViT+bs64+224	ResNet18 / 128 ² / grad accum
Transfer “fails”	Comparing joint-space policies	Ensure EE-relative + wrist obs
Ingest drops all frames	Kin-filter wrong URDF	Loosen limits; log reject reason
Wild ckpt fails outdoor	Direct sunlight (README known)	Document domain; don’t overclaim

What not to invent / claim

- Do not invent GPU wall-clock hours (paper omits them — cite epochs × GPU counts only).
- Do not claim external third-person cams in UMI policy (wrist-only).
- Do not equate UMI collection with DROID robot-in-wild teleop.
- Do not claim “70% OOD” without the 43/60 breakdown.
- Do not claim ACT drop-in is empirically equal (stated, **not** ablated).

Empiricist exam bite

“UMI = handheld wild teaching **without** robots on-site; relative EE **100%** vs abs **25%**; latency matching turns toss **87.5% – 57.5%**; fisheye **100% – 55%**; wild **1400/30 → 71.7%** vs narrow **0%**; DP **Ta=6 @10 Hz**; smoke RC0 → RC2 for directional proofs — never invent GPU hours.”

6.2 Scientific Reviewer — WEIGHT 4 (substantial)

Claim under review: Careful handheld interface + latency-matched relative-trajectory Diffusion Policy enables zero-shot transferable dynamic/bimanual/long-horizon skills from in-the-wild human demos.

Strengths

1. **Systems completeness:** names four prior failure modes (FoV, SfM precision, latency, multimodal capacity) and closes each with concrete design (HD/PD).
2. **Ablations that bite:** fisheye, mirrors+reflect, relative vs abs/delta, latency, inter-gripper, ViT vs ResNet — large effect sizes, matched initials (Ablation Bot ranks #1–#8).
3. **Cross-embodiment:** same cup ckpt UR5 → Franka **90%** with honest joint-limit fails.
4. **OOD data lesson:** ViT-L + narrow data = **0%**; diversity required — antidote to “just finetune CLIP.”
5. Open hardware+software sufficient for third-party reproduce (SLAM difficulty disclosed).
6. Correctly builds on **Diffusion Policy** rather than inventing a new generative head.
7. Fair matched initial-state protocol across methods.

Weaknesses / threats

1. **Small eval n** (often 20 episodes); operator-judged success — videos help, stats thin; no multi-seed train variance.
2. **SLAM texture dependence** admitted; wild success conditioned on mappable scenes.
3. **Kinematic filtering** pushes embodiment limits to post-hoc data drop, not embodiment-aware learning.
4. Still slower than bare hand; bulk/weight (**780 g**) limits marathon collection.
5. Parallel-jaw only — no dexterous-hand transfer story.
6. Latency numbers are **per-install**; Fig. 5's 100/120 ms are examples, not universals.
7. “~70% OOD” blends train/test cups and two scenes — report **43/60** precisely.
8. Absolute-action baseline may be a strawman under oracle calib — paper's point is calib-free deploy, so report clean **and** noisy-origin conditions.
9. ACT-as-drop-in claimed but **not empirically compared** — TENTATIVE if asserted equal.
10. ViT vs ResNet on dish: architecture × pretrain × task difficulty entangled.
11. Object-level 105/120 tossing vs episode-level elsewhere — keep metric defs straight.
12. Confounded “system” win: hard to isolate hardware vs action repr vs latency vs DP capacity — Empiricist RC1/RC2 correctly hold data fixed.

Verdict: Strong **systems + data-interface** paper; empirics are real-robot and ablation-rich. Main risk is overgeneralizing from cup-centric OOD and SLAM-friendly scenes. Accept-level RSS systems contribution.

Reviewer exam bite: Cite **relative 100/80/25** and **latency –30 pp** before claiming “UMI just works”; demand **43/60** not “~70%”; punish “ACT equals DP” without an ablate.

6.3 Visionary — WEIGHT 2 (substantial; BEHAVIOR + follow-ups; no Discord/Drive)

BEHAVIOR connection:

UMI is a **data-engine** prior for household long-horizon stacks that BEHAVIOR-class benchmarks stress (wash, fold, multi-step kitchen). Relative wrist-centric EE + DP/FM chunks align with π_0 / $\pi_{0.5}$ / **OpenPI** control habits; $\pi_{0.5}$ (2504.16054) and **GR00T-N1** (2503.14734) cite UMI (**CLEAR** from local fulltexts). For 2026 BEHAVIOR, UMI-style **handheld scaling** is the complementary pole to **DROID robot-in-wild** and **ALOHA lab teleop** — useful when organizers need **cheap scene diversity** without shipping Frankas.

2026 BEHAVIOR handheld / in-the-wild teaching recipe (steal the recipe, not necessarily the GoPro BOM):

1. **Teach anywhere:** prioritize wrist-centric portable demonstrators over lab-only teleop for scene diversity.
2. **Standardize the interface, not the arm:** relative EE (+ continuous gripper) + wrist FoV + **latency sheet** per deployment — then swap UR/Franka/mobile bases.
3. **Latency card mandatory:** every BEHAVIOR deploy publishes measured obs/infer/exec latencies and whether matching is on — especially for dynamic tasks.
4. **Diversity cards:** log scene / object / demonstrator ids (Fig. 9); run Empiricist matched-N wild vs narrow ablations before claiming generalization.
5. **Compose with fleet data:** mix UMI-style hand-held shards with DROID/OXE robot shards under schema tags (`source=handheld|robot` , `action=rel_ee` , `latency_matched=bool`).
6. **Race on modern policies:** keep UMI interface; swap DP → $\pi_{0.5}$ / **OpenPI** / **N1-class** FM heads with Empiricist horizon×latency grid.
7. **Long-horizon + recovery:** dish-style stage SR and mid-episode perturbation recovery = first-class BEHAVIOR metrics.
8. **Honest non-goals:** do not require reproducing OptiTrack SLAM ATE or 4 real tasks in course smoke; require RC0–RC2 directional proofs + written latency/action card.

Follow-up research

1. Embodiment-aware policies that consume kinematically-invalid demos (paper limitation #1).
2. Textureless-room action recovery (third-person + gripper fiducials).
3. Lighter / higher-DoF handheld (dex hand) while keeping observation alignment.
4. Swap DP → **flow matching** (π_0 -style) under same UMI interface; keep latency matching.
5. RTC / soft-inpaint async chunks on top of UMI's latency model for high-Hz arms.
6. Mobile manipulator + UMI relative EE (paper already moves base mid-rollout).
7. Internet-scale distributed MP4 collection (§IX vision) with SLAM QA filters.

New applications

- Rapid skill cloning in homes/restaurants without robot install during teach.
- Cross-robot skill libraries (one demo set → UR/Franka/ARX).
- Dynamic sports-like tossing/sorting beyond quasi-static BC.
- Bimanual deformable + articulated appliance tasks as stress tests for VLA finetunes.

Visionary one-liner: UMI turns “in-the-wild” from **robot logistics** into **human logistics** — the missing portable **action-rich** demo pipe that later VLAs still cite when they need wrist-aligned, latency-honest manipulation data.

Visionary exam bite: “For BEHAVIOR’26, steal UMI’s **relative EE + latency card + handheld scene diversity**; race on $\pi_{0.5}$ /N1 — don’t retrain 2024 ResNet DP as the entry; $\pi_{0.5}$ /GR00T **CLEAR** cite UMI.”

(No Discord / Drive checklist content.)

6.4 Stakeholder (produced despite empty seat)

Product one-liner: Open **GoPro gripper** → **Diffusion Policy** pipe that buys **action-diverse in-the-wild** demos without wheeling a robot into the cafe.

Who cares

- Labs wanting **wild visual diversity** without Franka fleets (vs DROID cost).
- Companies shipping policies across **multiple arms** (relative EE + twin wrist cam).

- Course / BEHAVIOR teams needing a **reproducible handheld BC** recipe.
- **Non-consumers**: no SLAM expertise; need dex hands; pure language-VLA without wrist cams.

Buy / build decision

1. Goal = **dynamic + bimanual BC from humans** → build UMI (~\$400 + printer) before teleop carts.
2. Goal = **76k Franka trajectories in real buildings** → DROID-style robot-in-wild still wins scale.
3. Goal = **fine bimanual lab skills** → ALOHA/ACT joint teleop still strong.
4. Always budget a **latency calibration day** on first deploy — not optional for toss-class tasks.

Risks: SLAM support burden; sunlight/domain shift; parallel-jaw ceiling; operator-eval success inflation; 780 g marathon fatigue.

Stakeholder exam bite: “UMI buys action-rich wild demos at ~\$400 without robot logistics; DROID buys scene-volume with robots; ALOHA buys lab bimanual precision — pick the pole, then budget the latency day.”

6.5 Archaeologist (produced despite empty seat)

Priors UMI absorbs (CLEAR)

- **Diffusion Policy** (Chi et al., 2303.04137) — generative action chunks + receding control [paper [9]].
- Handheld quasi-static grippers: Grasping-in-the-Wild [41], Visual Imitation Made Easy [50], Dobb-E [36] — UMI’s foil for **action poverty**.
- **ORB-SLAM3** + Urban GoPro/IMU calibration stack.
- Soft/series-elastic fingers (SEED / Alspach TRI).
- ACT [53] named as potential drop-in (**not** ablated).

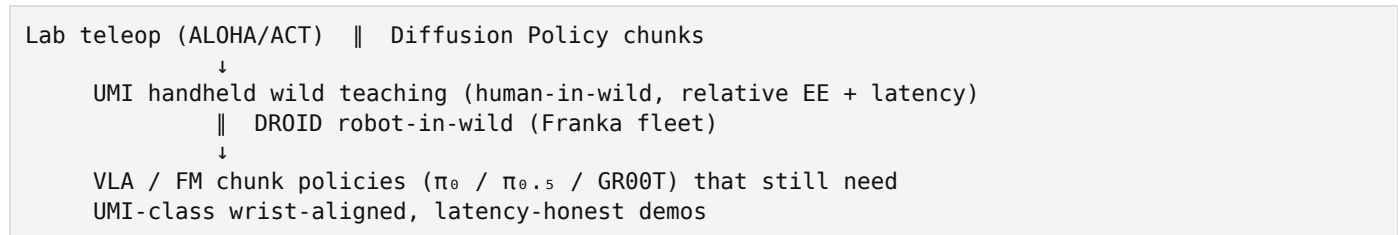
Contemporaneous contrast

- **ALOHA/ACT** lab puppet teleop (joint-space, embodiment-specific).
- **DROID** (Mar 2024, close arXiv neighbor) robot-in-wild Franka teleop — opposite “who goes into the wild.”
- Mobile ALOHA / GELLO — still robot-present collection.

Descendants / citations (CLEAR from local packs/fulltexts)

- $\pi_{0.5}$ (2504.16054) bibliography cites UMI.
- **GR00T-N1** (2503.14734) cites UMI.
- Downstream deploy: umi-on-legs, umi-arx; community handheld forks.
- Conceptual line into wrist-cam + relative-EE + chunk policy stacks (OpenPI recipes, BEHAVIOR finetunes) — **TENTATIVE** as named default baseline.

Genealogy



Genealogy one-liner: Lab teleop || DP chunks → **UMI handheld wild teaching** || DROID robot-in-wild → VLA/FM policies that still cite UMI for wrist-aligned demos.

Archaeologist exam bite: “UMI = in-the-wild **without** in-the-wild robots; relative EE + latency matching + fisheye wrist; DP Ta=6; OOD cup ~72% after 1400/30; $\pi_{0.5}$ /GR00T **CLEAR cite**.”

7. Empiricist run pack (env / scripts / GPU — from DELIVERABLE)

Env setup

```
export UMI_WORK_ROOT=$HOME/src
export UMI_DATA_ROOT=$HOME/data/umi_subsets    # EDIT – subsets / synthetic OK
cd /workspace/umi-quiet
bash env_outline/setup_mamba.sh
# mamba activate umi-scout
mkdir -p logs
# EDIT: clone official UMI + Diffusion Policy; pin commits → logs/pins.txt
# EDIT partition/account in slurm_*.sh
# Outlines only – do NOT sbatch from quiet box
```

File	Role
env_outline/conda_env.yaml	umi-scout sketch (torch + DP/UMI deps placeholders)
env_outline/setup_mamba.sh	Create/update env; data/log dirs
env_outline/run_checklist.md	Ordered RC0 → RC4 checklist
env_outline/slurm_demo_ingest_smoke.sh	CPU/light ingest gate
env_outline/slurm_relative_action_ablation.sh	Array over {rel,abs,delta}
env_outline/slurm_latency_sweep.sh	Eval-only latency grid
env_outline/slurm_policy_train_smoke.sh	1-GPU short DP train
env_outline/slurm_embodiment_transfer_stub.sh	IK/feasibility cross-robot

GPU / RAM / TIME budget honesty

Assumptions: scout DP (CNN or small ViT), image ≤ 224 , batch 32–64, **not** full paper real-robot protocol. **Wall-clock hours for paper full trains are NOT reported — do not invent them.**

Setup	GPU VRAM (peak)	Host RAM	Time / run (order)
RC0 ingest (50 eps)	—	8–16 GB	<0.5 h CPU
RC1 action-repr ×3 short	10–18 GB	32 GB	2–8 h total A100
RC2 latency sweep (infer)	same as ckpt	16–32 GB	<1–4 h
RC3 policy smoke 5k–20k	12–24 GB (ViT higher)	32–64 GB	4–12 h A100; ~1.5–2× on 4090
RC4 embodiment stub	≤RC3 or CPU	16 GB	<1–3 h
Dish-like ViT FT stretch	20–40 GB	64 GB	8–20 h — defer if tight
Paper full 4-task real eval	robot time + multi-day train	—	Out of course smoke

Course envelope: target $\leq \sim 25$ GPU-h for P0–P1 matrix (RC0–RC4). Full HD/PD × multi-seed × real robots = **hundreds of GPU-h + robot hours** — do not promise.

Deploy latency budget: paper-scale arm/gripper exec $\sim 100\text{--}120$ ms examples; DP infer (DDIM×16) must leave room inside control period after matching — link </workspace/diffusion-policy-quiet/> .

Success criteria vs paper claims

Claim (paper)	Minimal course criterion	Stretch
Relative $\gg$ Absolute (100% vs 25%)	RC1: Rel beats Abs on origin-jitter OOD	Match clean-ID ranking vs Delta
Relative $\geq$ Delta (100% vs 80%)	Rel $\geq$ Delta on ID MSE / proxy	Full cup-proxy success
Latency matching critical (~ 30 pp)	RC2: match $>$ zero on dynamic velocity/timing	Quantify $-pp$ on real toss
Cross-embodiment 90%	RC4: feasibility transfer with kin-filter	Real Franka 18/20
Inter-gripper (70% vs 30%)	Toggle improves sync on dual-arm shard	Real cloth 14/20
In-the-wild $\sim 70\%$ OOD	Multi-scene $>$ single-scene @ matched N	Cafe/fountain protocol
DP fits multimodal	RC3 samples both rotate modes if present	Cup 20/20
SLAM ATE ~ 6.1 mm / 3.5°	Ingest reports pose quality proxy	Reproduce SVII on hardware
3 $\times$ teleop throughput	Out of scope (hardware study)	—

Pack pass bar: Empiricist can execute RC0–RC2 from outlines and obtain **paper-aligned directional** results on proxies; RC3–RC4 attempted or blocked only by missing public shards (document blocker).

8. Slide Maker brief

Role	Weight / timing guide	Focus
Empiricist	1 · LONGEST	Hypotheses, RC0–RC4, exact reproduce cmds, GPU honesty, ablation matrix — ~ 16–18 slides
Reviewer	4 · substantial	Ablations that bite; 43/60 not “ $\sim 70\%$ ”; verdict Accept systems
Visionary	2 · substantial	BEHAVIOR handheld recipe; $\pi_{0.5}/GR00T$ cite; no Discord/Drive
Stakeholder	filled (empty seat)	Buy/build poles vs DROID/ALOHA; latency day
Archaeologist	filled (empty seat)	Genealogy ALOHA $\parallel$ DP $\rightarrow$ UMI $\parallel$ DROID $\rightarrow$ $\pi_{0.5}/GR00T$

Spine (paste into Slide Maker) — 5 bullets:

- In-the-wild teaching without in-the-wild robots** — human logistics $\rightarrow$ robot-ready demos via relative EE + latency honesty.
- Must-cite numbers:** cup **100%/90%**; toss **87.5%** (no lat **57.5%**); wild **1400/30** $\rightarrow$ **71.7%** vs narrow **0%**; rel **100%** vs abs **25%**; fisheye **100%** vs **55%**.
- Motion:** DP $T_a=6$ @10 Hz (toss 20 Hz); relative SE(3) w.r.t. chunk (t_0); latency match (discard stale + send-ahead; $\sim 100/120$ ms examples).
- Collection:** $\sim \$371$ GoPro gripper; map-then-localize SLAM; $>3\times$ teleop throughput; complementary to **DROID** (robot-in-wild) and **ALOHA** (lab teleop).
- BEHAVIOR’26 + lineage:** steal relative EE + latency card + handheld diversity; race on $\pi_{0.5}/N1$; $\pi_{0.5}/GR00T$ **CLEAR** cite UMI; Empiricist RC0–RC2 directional smokes — **never invent GPU hours**.

GitHub title sketch: [ECE 605] - UMI Making Empiricist

Full Slide Maker brief: /workspace/umi-quiet/SLIDE_BRIEF.md

Paths

Artifact	Path
This pack	/workspace/umi-quiet/KANTA_PACK.md
PDF	/workspace/umi-quiet/KANTA_PACK.pdf
Slide brief	/workspace/umi-quiet/SLIDE_BRIEF.md
Senior Research	/workspace/umi-quiet/DELIVERABLE.md
Summarizer	/workspace/papers/umi-2402.10329-summary.md
Ablations	/workspace/papers/umi-2402.10329-ablations.md
Paper notes	/workspace/umi-quiet/paper_notes.md
Env outlines	/workspace/umi-quiet/env_outline/
Paper PDF	/workspace/papers/umi-2402.10329.pdf
Page figs	/workspace/papers/umi-figs/page-*.png

Pack synthesized 2026-09-29 PT (HST) for KanBot / ECE 605 Paper Manager — Nov 16 Data · Empiricist=1 (Kanta PRIMARY) · Reviewer=4 · Visionary=2 · all five seats produced.